		A	B	C	D	E	F
Rung #1	1	CNC operation stops when receiving a signal from the vision sensor					
	2			Bit Vision_signal			Bit CNC_op reset
	3						
Rung #2	4	Operation is halted and tool is removed from the CNC machine					
	5			Bit CNC_op			Bit Tool_remove out
	6						
Rung #3	7	The tool being removed activates the tool conveyor, bringing the old tool to a "used tool" palette while bringing a new tool from a "new tool" palette.					
	8			Bit Tool_remove			Bit Tool_Carousel out
	9						
Rung #4	10	A sensor placed adjacent to the rear end of the CNC machine detects a new tool coming from the other side and triggers a timer to create a delay. This is so the tool reaches the slot instead of stopping when in front of the sensor.					
	11			Bit CNC_sensor			Start Timer i32 Insert_timer
	12						
Rung #5	13	The conveyor remains active for a small interval before stopping so the tool reaches the proper position.					

		A	B	C	D	E	F
Rung #5	14			Insert_timer (i32) < 0.3 sec			Bit Tool_Carousel — reset
	15						
Rung #6	16	The timer also creates a delay between the tool arriving at the position and entering the slot in the CNC machine					
	17			Insert_timer (i32) < 2 sec			Bit Tool_insert — out
	18						
Rung #7	19	With a fresh tool in the slot, the machining operation can now resume!					
	20			Bit Tool_insert			Bit CNC_op — out
	21						
Rung #8	22	Resetting the tool's stationary timer once the operation resumes					
	23			Bit CNC_op			Pause — Timer: i32 Insert_timer
	24						
Rung #9	25	When the new tool palette is empty, it sets off an alarm light to indicate that it needs a new batch of fresh tools. Since the palettes for new and old tools are of the same size, the operator can also use this light to dispose of all the damaged tools in the other palette as well.					
	26			Bit Palette_sensor			Bit Palette_empty — out

Rung

		A	B	C	D	E	F
	27						
Rung #10	28						no-op
	29						
	30						
	31						
	32						
	33						
	34						
	35						
	36						
	37						
	38						
	39						